

Algorithms and Data Structures

Problem Sheet 2

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Remark Due to the technical difficulties in the Wednesday lecture, I did not have access to my personal lecture preparation, so I used the chapter in the book from the library. Unfortunately, what I forgot is that there is a slight error in the book concerning the definition of the polynomial growth condition. The correct definition requires a different order of the quantifiers.

We say that a function $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies the polynomial growth condition if

$$\forall \varphi > 1 : \exists d > 1 : \exists N > 0 : \forall n \geq N : \forall 1 \leq \psi \leq \varphi : \frac{1}{d}f(n) \leq f(\psi n) \leq df(n).$$

Note the swap of the $\exists N > 0$ and the $\forall \varphi > 1$.

18. Use the Master Theorem to provide Θ -Bounds for the following recursions (assume $n_0 = 1$ for all of them).
 - (a) $T(n) = 2T(\frac{n}{4}) + \Theta(1)$,
 - (b) $T(n) = 2T(\frac{n}{4}) + \Theta(\sqrt{n})$,
 - (c) $T(n) = 2T(\frac{n}{4}) + \Theta(\sqrt{n} \lg^2 n)$,
 - (d) $T(n) = 2T(\frac{n}{4}) + \Theta(n)$,
 - (e) $T(n) = 2T(\frac{n}{4}) + \Theta(n^2)$.
19. Professor Caesar wants to develop a matrix-multiplication algorithm that is asymptotically faster than Strassen's algorithm. His algorithm will use the divide-and-conquer method, dividing each matrix into submatrices of size $\frac{n}{4} \times \frac{n}{4}$, and the divide and combine steps together will take $\Theta(n^2)$ time. Suppose that the professor's algorithm creates a recursive subproblems of that size. What is the largest integer value of a for which his algorithm could possibly run asymptotically faster than Strassen's? How does a change if the subproblems have size $\frac{n}{3} \times \frac{n}{3}$ or $\frac{n}{5} \times \frac{n}{5}$?
20. Consider the function $f(n) = \lg n$. Argue that although $f(\frac{n}{2}) < f(n)$, the regularity condition $af(\frac{n}{b}) \leq cf(n)$ with $a = 1$ and $b = 2$ does not hold for any constant $c < 1$. Argue further that for any $\varepsilon > 0$, the condition in Case 3 of the Master theorem that $f(n) = \Omega(n^{n \log_b a + \varepsilon})$ does not hold.
21. Show that for suitable constants $a > 0$, $b > 1$, $\varepsilon > 0$ the function $f(n) := 2^{\lceil \lg n \rceil}$ satisfies all conditions of Case 3 of the Master theorem except for the regularity condition.

22. Show that $f(n) = n^2$ satisfies the polynomial growth condition but $g(n) = 2^n$ does not.
23. Prove that the polynomial growth condition is equivalent to the following condition on the function $f : \mathbb{R} \rightarrow \mathbb{R}$:
- For all $0 < \varphi < 1$ there exist constants $N > 0$ and $c_1, c_2 > 0$ with the property that for all $n \geq N$ and all $u \in [\varphi n, n]$

$$c_1 f(n) \leq f(u) \leq c_2 f(n).$$

24. Prove that $f(n) = n \lg n$ and $f(n) = n \lg \lg n$ satisfy the polynomial growth condition.
25. Assume that f satisfies the polynomial growth condition, and let $g(n) = \Theta(n)$. Show that $f(g(n)) = \Theta(f(n))$. Usually we write $f(\Theta(n)) = \Theta(f(n))$. Explain why this notation makes sense.
26. Use the Akra-Bazzi-Theorem to compute Θ -Expressions for the following recurrences:
- (a) $T(n) = T(\frac{n}{2}) + T(\frac{n}{3}) + T(\frac{n}{6}) + \Theta(n \lg n)$,
 - (b) $T(n) = 3T(\frac{n}{3}) + 8T(\frac{n}{4}) + \Theta(\frac{n^2}{\lg n})$,
 - (c) $T(n) = \frac{2}{3}T(\frac{n}{3}) + \frac{1}{3}(\frac{2n}{3}) + \Theta(\lg n)$,
 - (d) $T(n) = \frac{1}{3}T(\frac{n}{3}) + \frac{1}{n}$,
 - (e) $T(n) = 3T(\frac{n}{3}) + 3T(\frac{2n}{3}) + n^2$.
27. Use the Akra-Bazzi-Theorem for finding another proof for the continuous Master Theorem.
28. Solving recurrences using a change of variables. Let us consider the recurrence $T(n) = 2T(\sqrt{n}) + \Theta(\lg n)$. For this recurrence the Master Theorem cannot be used directly. Therefore, define $m = \lg n$ and $S(m) = T(2^m)$. Rewrite the recurrence for T in n into a recurrence for S in m . Use the Master Theorem to solve the recurrence for S and rewrite the solution in terms of T and n .
29. Use the method of problem 28 to solve the following recurrences:
- (a) $T(n) = 2T(\sqrt{n}) + \Theta(1)$,
 - (b) $T(n) = 3T(\sqrt[3]{n}) + \Theta(n)$,
 - (c) $T(n) = 3T(\sqrt{n}) + 3T(\sqrt[3]{n}) + \Theta((\lg \lg n)^2)$.